Supplemental tables and figures

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This document includes all the supplemental tables and figures that accompany "Precipitation moderates associations between native shrubs and invasive buffelgrass (*Pennisetum ciliare*) growth in the Sonoran Desert while local topography has less influence" by Ossanna et al. (2026).

Tables

Table S1. Actual values averaged by year

Actual values of variables measured, averaged across all plots for each year (to contextualize year-to-year change values used in analyses). Precipitation measured in mm, buffelgrass density measured in individuals per m^2, and cover measured in percent.

		Buffelgrass				Native vegetation	
Year	Precipitation	Total culms	Repro culms	Density	Cover	Shrub cover	Herb cover
2020	222	24 ± 19	12 ± 13	3.8 ± 1.5	39 ± 19	18 ± 12	5.9 ± 10.5
2021	362	38 ± 33	14 ± 16	8 ± 7.6	40 ± 21	13 ± 8	9.9 ± 8.7
2022	264	53 ± 40	26 ± 25	6.8 ± 4.1	35 ± 17	12 ± 10	5.1 ± 6.4
2023	250	129 ± 80	53 ± 35	3 ± 1.5	39 ± 13	22 ± 13	2.6 ± 2.7

Table S2. Total culms model

Model coefficients of the buffelgrass culm count model (response variable: year-to-year change in total number of culms). Note that the reference level for Aspect is east (E). Marginal $R^2 = 0.128$; conditional $R^2 = 0.416$.

Coefficient	Estimate	Std. Error	df	t value	Pr(>|t|)
Intercept	10.6052	10.4895	4.0032	1.011	0.36915
Previous year precipitation	-3.4630	1.5748	1199.7711	-2.199	0.02807*
Aspect, N	8.8074	7.8770	21.0337	1.118	0.27613
Aspect, S	3.8059	5.7441	21.5339	0.663	0.51464
Aspect, SW	17.1407	7.7785	20.8623	2.204	0.03893*
Aspect, W	6.7448	5.5320	20.2762	1.219	0.23675
Plot slope	-2.8999	1.4273	137.6064	-2.032	0.04411*
Change in BG density	-8.0743	1.4346	1135.2657	-5.628	2.29e-08***
Change in shrub cover	5.6466	1.1347	1202.4446	4.976	7.43e-07***
Change in herb cover	0.8591	1.1423	1177.3904	0.752	0.45213
Initial BG density	-3.3092	1.4710	183.8740	-2.250	0.02566*
Initial shrub cover	0.1346	1.1756	1024.3192	0.115	0.90886
Init herb cover	1.9152	1.4336	293.3460	1.336	0.18258
Change in BG density * Previous year precip	3.6112	1.6610	268.1289	2.174	0.03057*
Change in shrub cover * Previous year precip	-6.7247	1.2327	1159.8062	-5.455	5.98e-08***
Change in herb cover * Previous year precip	2.9944	1.1164	807.6179	2.682	0.00746**

Table S3. Reproductive culms model

Model coefficients of the buffelgrass reproductive culm count model (response variable: year-to-year change in number of reproductive culms). Note that the reference level for Aspect is east (E). Marginal $R^2 = 0.100$; conditional $R^2 = 0.305$.

Coefficient	Estimate	Std. Error	df	t value	Pr(>|t|)
Intercept	10.52774	5.02894	4.08865	2.093	0.10292
Previous year precipitation	-0.78486	0.94613	1122.28311	-0.830	0.40697
Aspect, N	4.53389	3.93244	16.69928	1.153	0.26517
Aspect, S	-3.28699	2.87943	17.35433	-1.142	0.26915
Aspect, SW	10.38755	3.88828	17.31964	2.672	0.01592*
Aspect, W	1.48971	2.75739	16.36345	0.540	0.59629
Plot slope	-0.03354	0.79615	70.88934	-0.042	0.96651
Change in BG density	-2.30663	0.85934	974.17164	-2.684	0.00739**
Change in shrub cover	1.50990	0.68213	1130.25569	2.213	0.02706*
Change in herb cover	2.91011	0.68657	1121.36034	4.239	2.43e-05***
Initial BG density	-1.79555	0.83026	100.41982	-2.163	0.03294*
Initial shrub cover	-0.42993	0.70080	781.71922	-0.613	0.53974
Initial herb cover	1.15812	0.82227	148.22307	1.408	0.16109
Change in BG density * Previous year precip	-1.59968	0.95139	148.25168	-1.681	0.09479
Change in shrub cover * Previous year precip	-0.76535	0.73938	1047.35479	-1.035	0.30085
Change in herb cover * Previous year precip	-0.41514	0.66024	520.26126	-0.629	0.52978

Table S4. Density model

Model coefficients of the buffelgrass density model (response variable: year-to-year change in density [individuals per m^2]). Note that the reference level for Aspect is east (E). Marginal $R^2 = 0.422$; conditional $R^2 = 0.468$.

Coefficient	Estimate	Std. Error	df	t value	Pr(>|t|)
Intercept	5.8557	1.5538	8.3902	3.769	0.00502**
Previous year precipitation	5.0587	0.7658	44.6092	6.606	4.07e-08***
Aspect, N	-2.6809	2.4598	12.6782	-1.090	0.29604
Aspect, S	-5.9883	1.9398	13.9153	-3.087	0.00808**
Aspect, SW	-3.3270	2.6377	15.2781	-1.261	0.22611
Aspect, W	-5.0737	1.8297	12.4658	-2.773	0.01636*
Plot slope	0.4289	0.6428	23.8655	0.667	0.51103
Change in shrub cover	1.3850	0.6742	102.9817	2.054	0.04248*
Change in herb cover	0.7413	0.7150	102.9794	1.037	0.30227
Initial BG density	-0.5874	0.6922	44.8344	-0.849	0.40063
Initial shrub cover	0.3926	0.6808	108.6825	0.577	0.56536
Initial herb cover	0.4361	0.7355	47.5811	0.593	0.55603
Change in shrub cover * Previous year precip	2.2715	0.7166	110.6371	3.170	0.00197**
Change in herb cover * Previous year precip	0.6214	0.6176	85.6731	1.006	0.31718

Table S5. Cover model

Model coefficients of the buffelgrass cover model (response variable: year-to-year-change in cover [%]). Note that the reference level for Aspect is east (E). Marginal $R^2 = 0.261$; conditional $R^2 = 0.285$.

Coefficient	Estimate	Std. Error	df	t value	Pr(>|t|)
Intercept	10.49784	3.82196	13.48579	2.747	0.016188*
Previous year precipitation	7.33466	1.99290	44.11544	3.680	0.000631***
Aspect, N	-6.99079	5.97868	22.87368	-1.169	0.254329
Aspect, S	-11.19915	4.72672	23.41969	-2.369	0.026427*
Aspect, SW	-9.98472	6.45741	26.73857	-1.546	0.133802
Aspect, W	-11.23305	4.43792	21.43132	-2.531	0.019246*
Plot slope	3.22815	1.59507	34.49457	2.024	0.050788
Change in shrub cover	-1.67363	1.77546	102.86844	-0.943	0.348071
Change in herb cover	2.09712	1.89206	109.30705	1.108	0.270131
Initial BG density	-0.88507	1.75434	61.44016	-0.505	0.615711
Initial shrub cover	-1.44194	1.77203	106.07076	-0.814	0.417627
Initial herb cover	2.22733	1.86387	57.83920	1.195	0.236963
Change in shrub cover * Previous year precip	0.69379	1.88062	109.88566	0.369	0.712901
Change in herb cover * Previous year precip	0.03172	1.60010	88.35380	0.020	0.984229

Table S6. Seedling survival model

Model coefficients of the buffelgrass seedling survival model (response variable: seedling survival [%]). Note that the reference level for Aspect is east (E). Marginal $R^2 = 0.493$; conditional $R^2 = 0.966$.

Coefficient	Estimate	Std. Error	z value	Pr(>|z|)
Intercept	-1.71172	0.80310	-2.131	0.0331*
Previous year precipitation	1.53652	0.08337	18.430	<2e-16***
Aspect, N	0.59861	0.40399	1.482	0.1384
Aspect, S	-0.43814	0.27521	-1.592	0.1114
Aspect, SW	0.52042	0.33425	1.557	0.1195
Aspect, W	0.16955	0.26274	0.645	0.5187
Plot slope	0.04097	0.07758	0.528	0.5975
BG density	0.45818	0.08564	5.350	8.8e-08***
Shrub cover	-0.15633	0.06397	-2.444	0.0145*
Herb cover	0.06656	0.06408	1.039	0.2990
BG density * Previous year precip	0.06006	0.08730	0.688	0.4915
Shrub cover * Previous year precip	-0.04669	0.05890	-0.793	0.4279
Herb cover * Previous year precip	-0.04339	0.06815	-0.637	0.5244

Figures

Figure S1

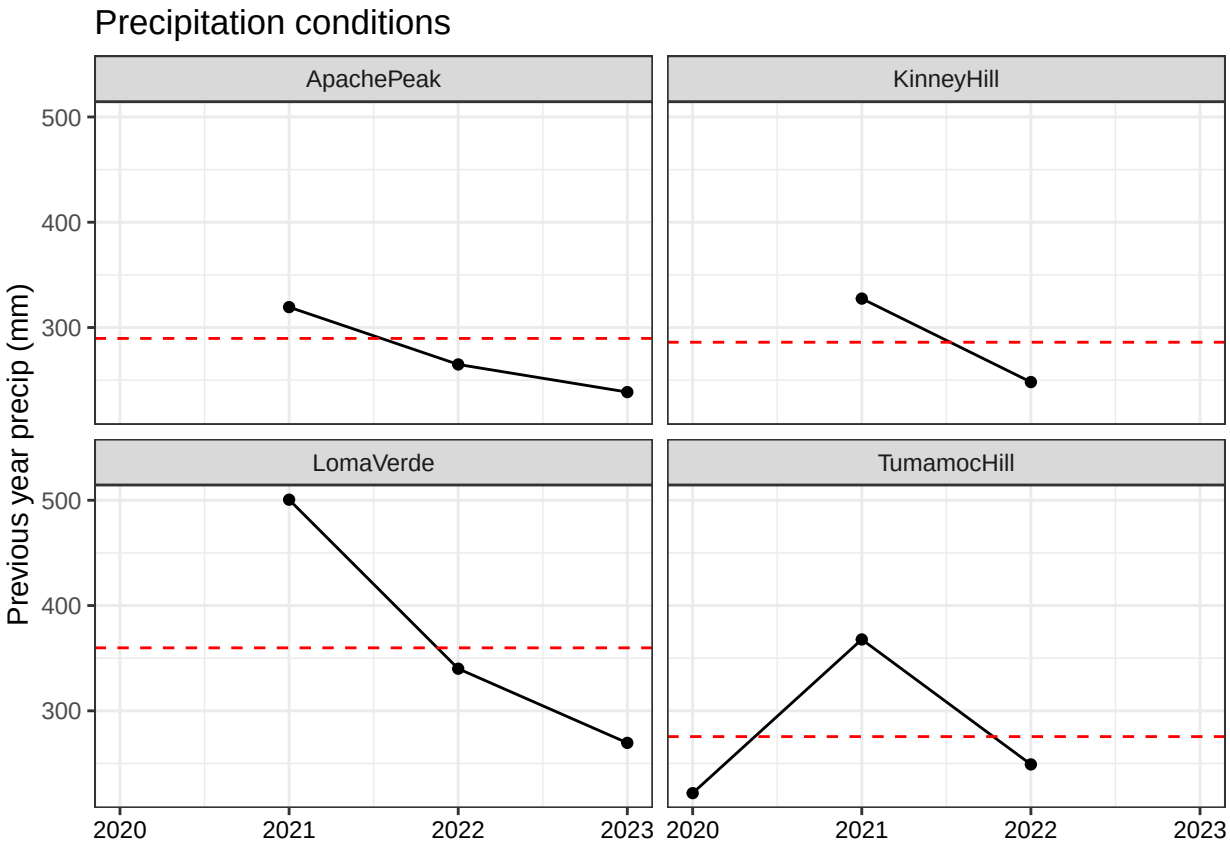

Figure S1. Precipitation conditions at each site during the study. Red dashed lines indicate mean annual precipitation of each site.

Figure S2

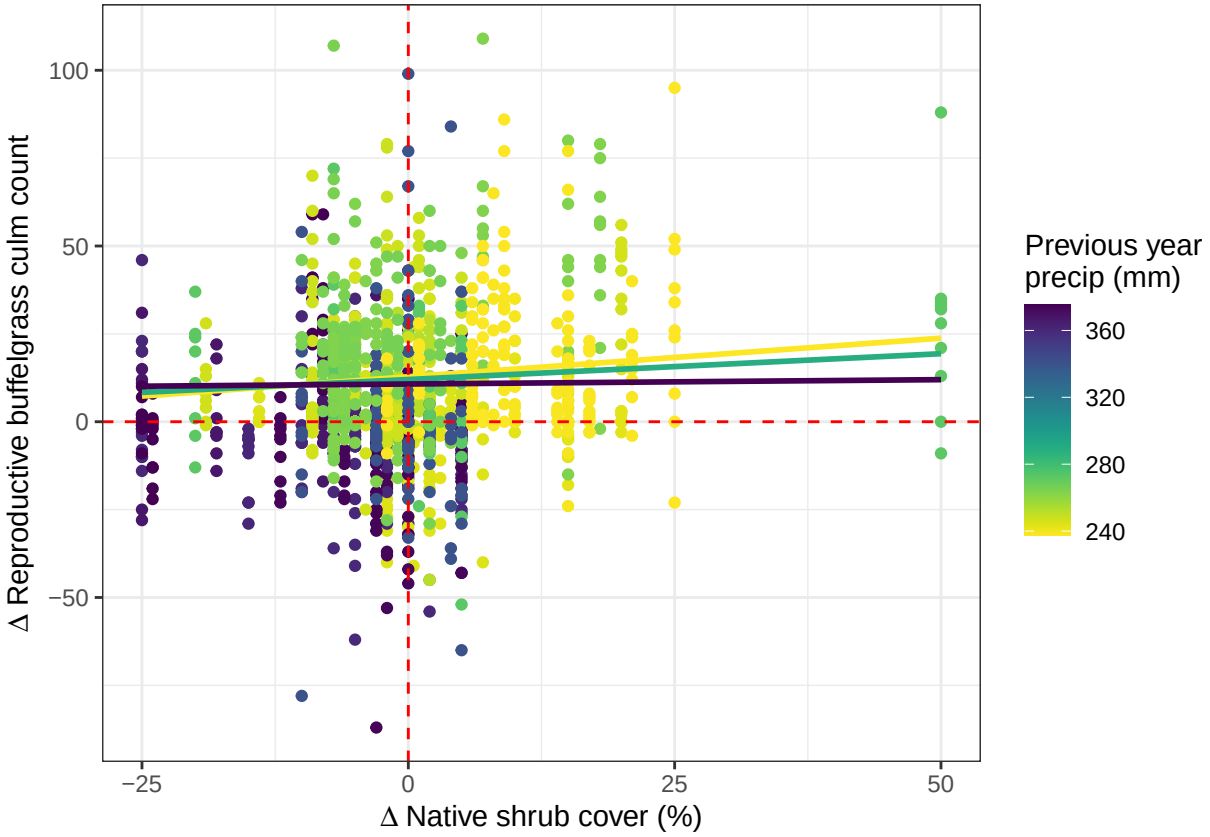

Figure S2. Relationship between year-to-year change in native shrub cover and year-to-year change in buffelgrass reproductive culm count, moderated by precipitation. Colored lines represent model-predicted values under the wettest precipitation conditions experienced (purple line), the driest conditions (yellow line), and average conditions (green line). Vertical red dashed line indicates no change in shrub cover while horizontal line indicates no change in the number of reproductive buffelgrass culms.

Figure S3

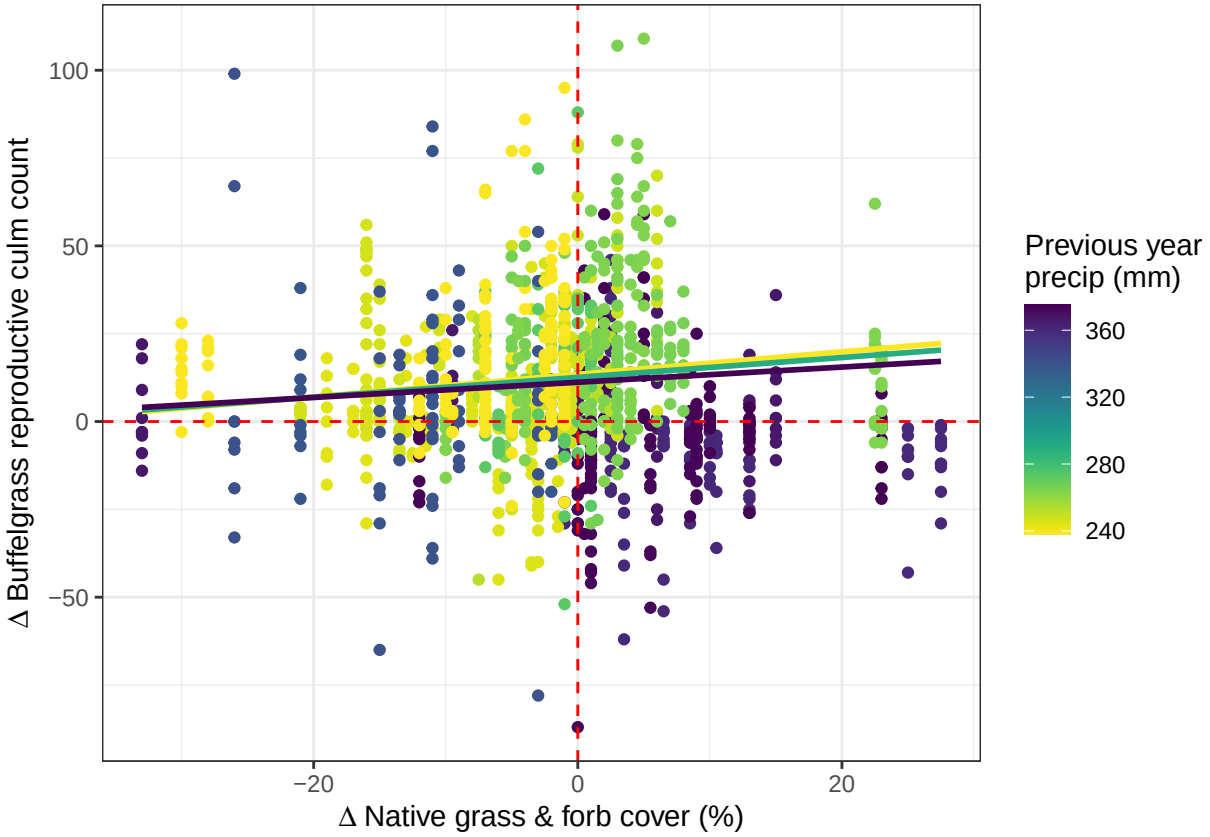

Figure S3. Relationship between year-to-year change in native herbaceous (grass and forb) cover and year-to-year change in buffelgrass reproductive culm count, moderated by precipitation. Colored lines represent model-predicted values under the wettest precipitation conditions experienced (purple line), the driest conditions (yellow line), and average conditions (green line). Vertical red dashed line indicates no change in herb cover while horizontal line indicates no change in the number of reproductive buffelgrass culms.

Figure S4

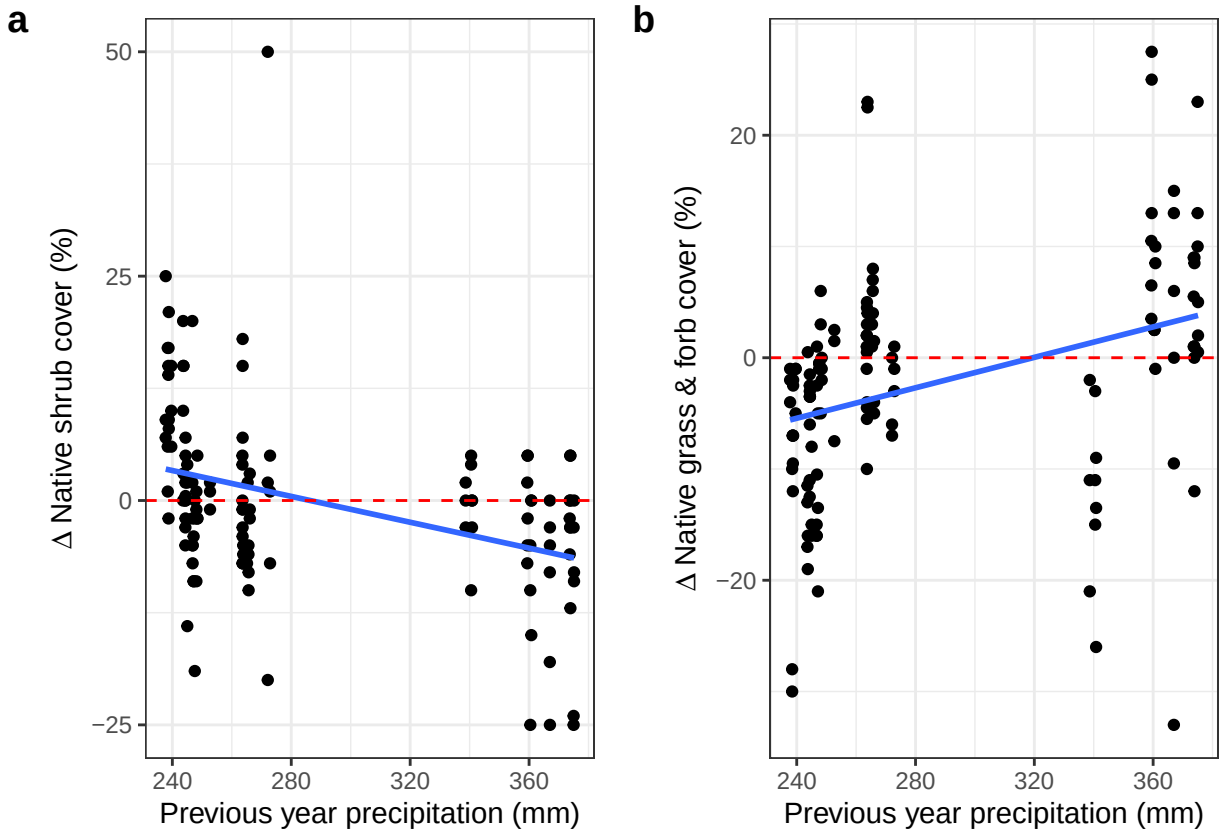

Figure S4. Simple linear regression of (A) year-to-year change in native shrub cover and (B) year-to-year change in native herbaceous cover as a function of the previous year's precipitation, pooling all sampling locations and times. Horizontal red dashed lines indicate no change in shrub or herbaceous cover.

Figure S5

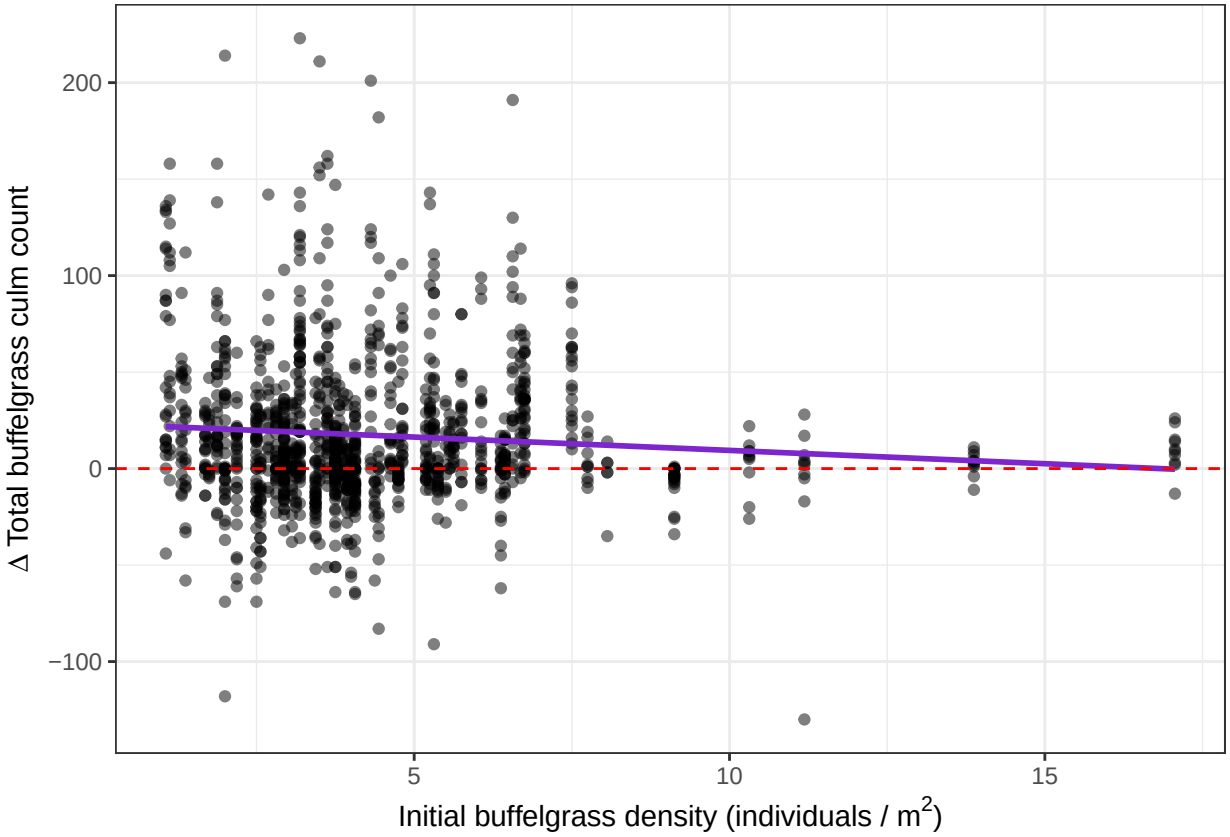

Figure S5. Relationship between initial buffelgrass density of the plot and year-to-year change in total buffelgrass culm count. Purple line represents model-predicted values. Horizontal red dashed line indicates no change in the number of buffelgrass culms.

Figure S6

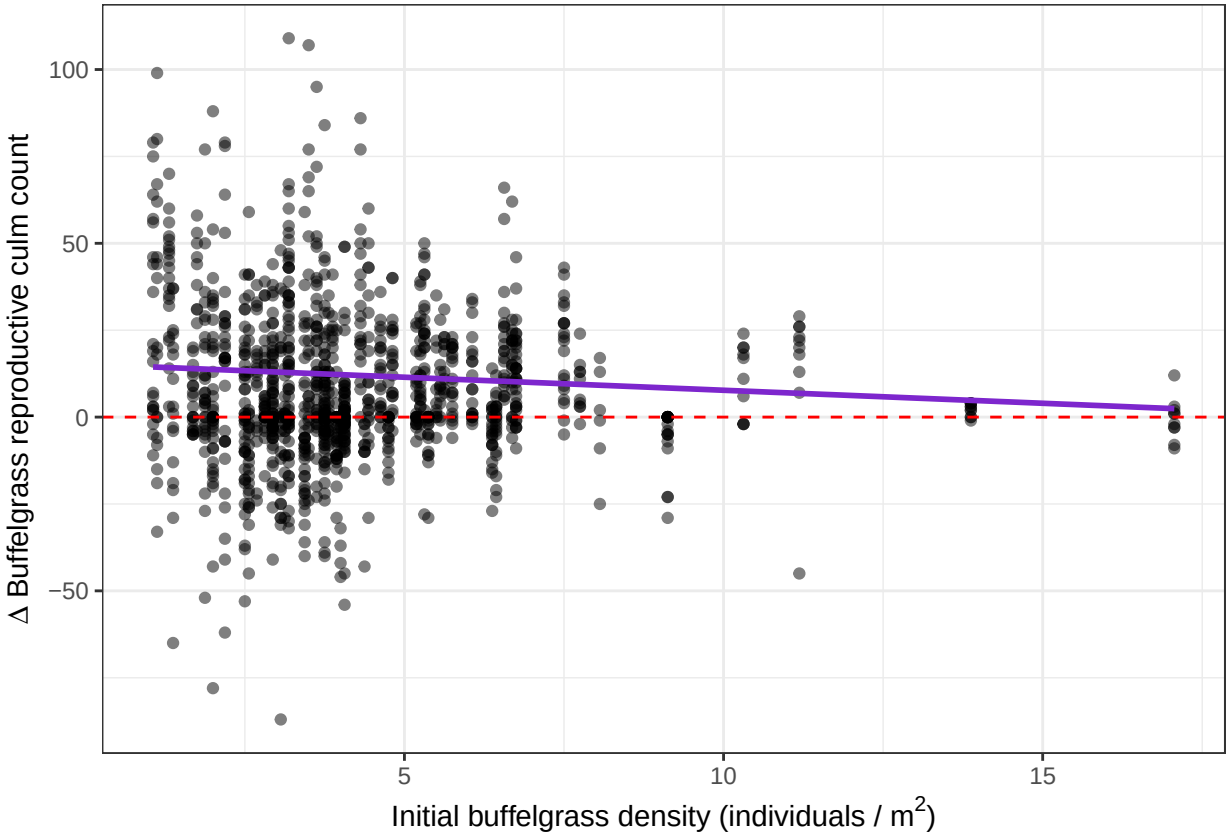

Figure S6. Relationship between initial buffelgrass density of the plot and year-to-year change in buffelgrass reproductive culm count. Purple line represents model-predicted values. Horizontal red dashed line indicates no change in the number of reproductive buffelgrass culms.

Figure S7

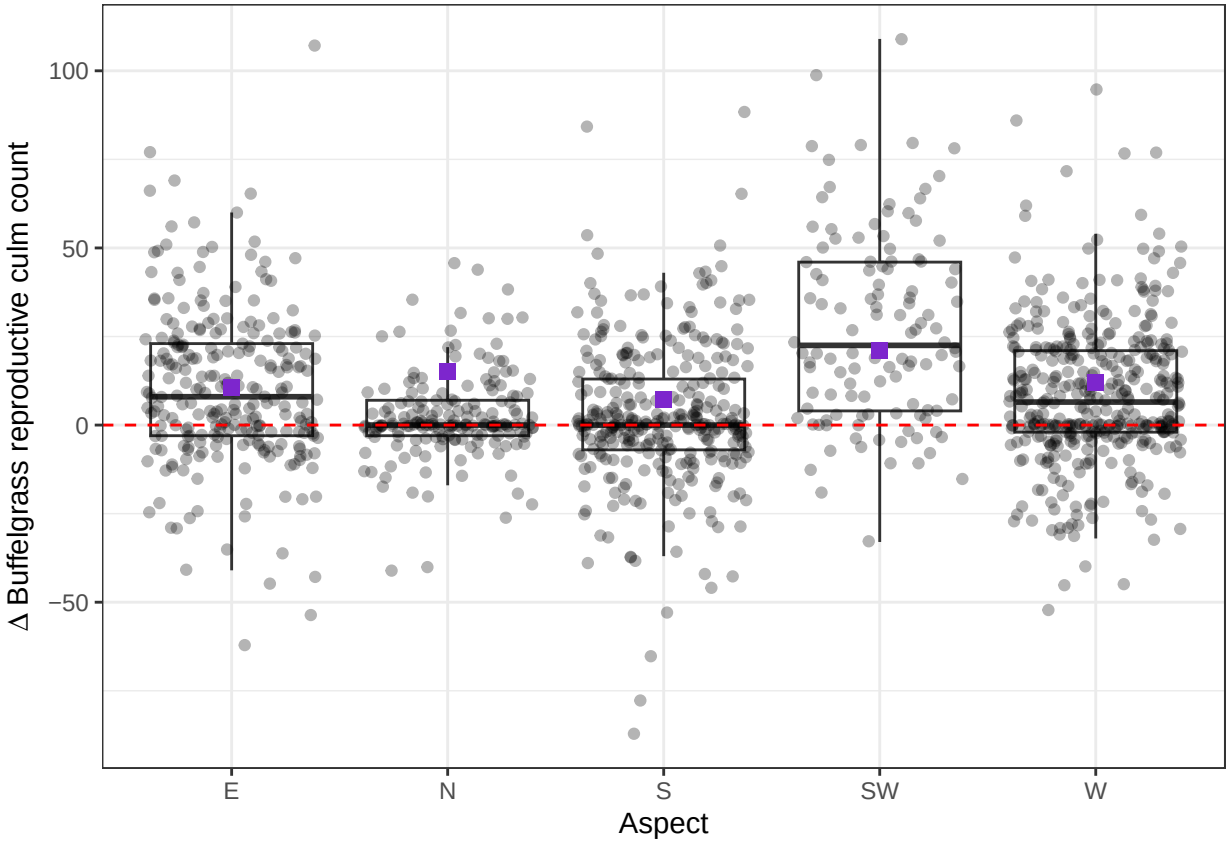

Fig. S7. Effects of aspect on year-to-year change in buffelgrass reproductive culm count. Purple squares represent model-predicted averages. Horizontal red dashed line indicates no change in the number of reproductive buffelgrass culms.

Figure S8

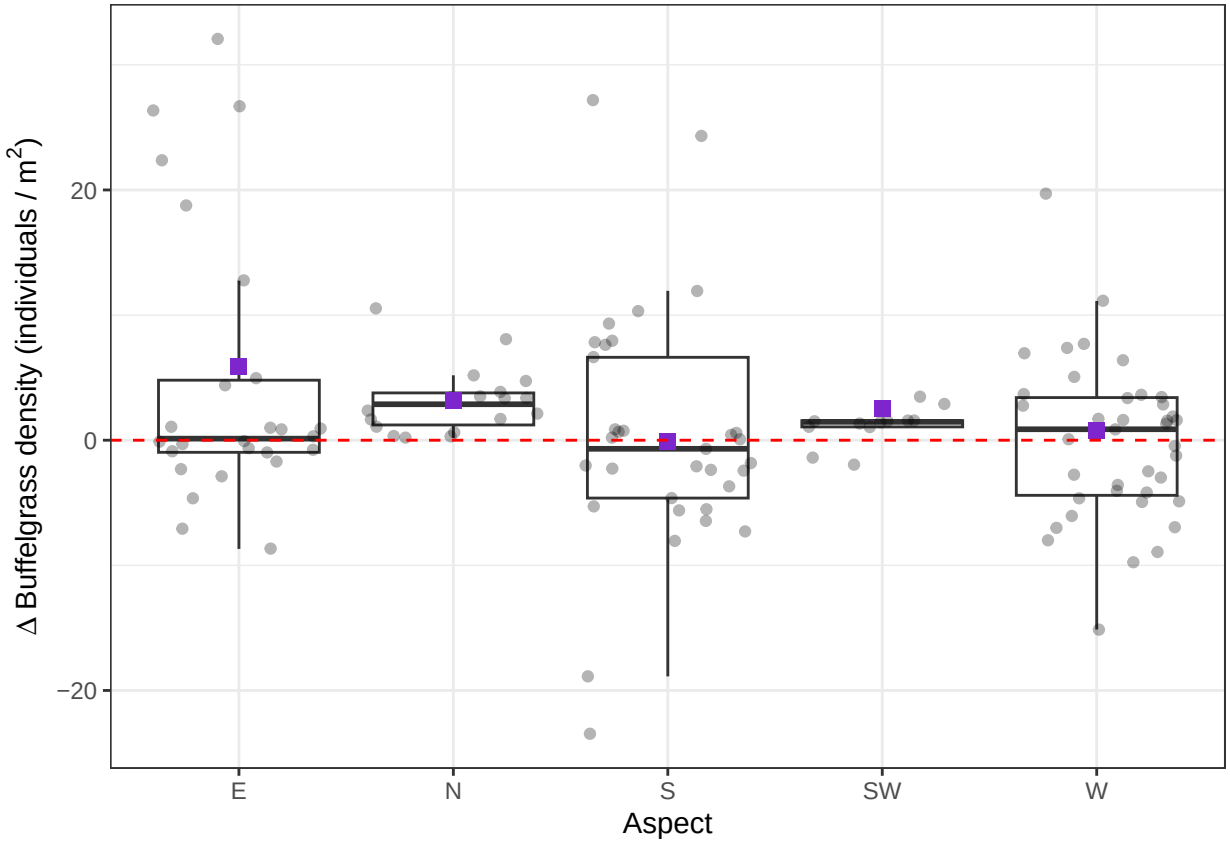

Fig. S8. Effects of aspect on year-to-year change in buffelgrass density. Purple squares represent model-predicted averages. Horizontal red dashed line indicates no change in buffelgrass density.

Figure S9

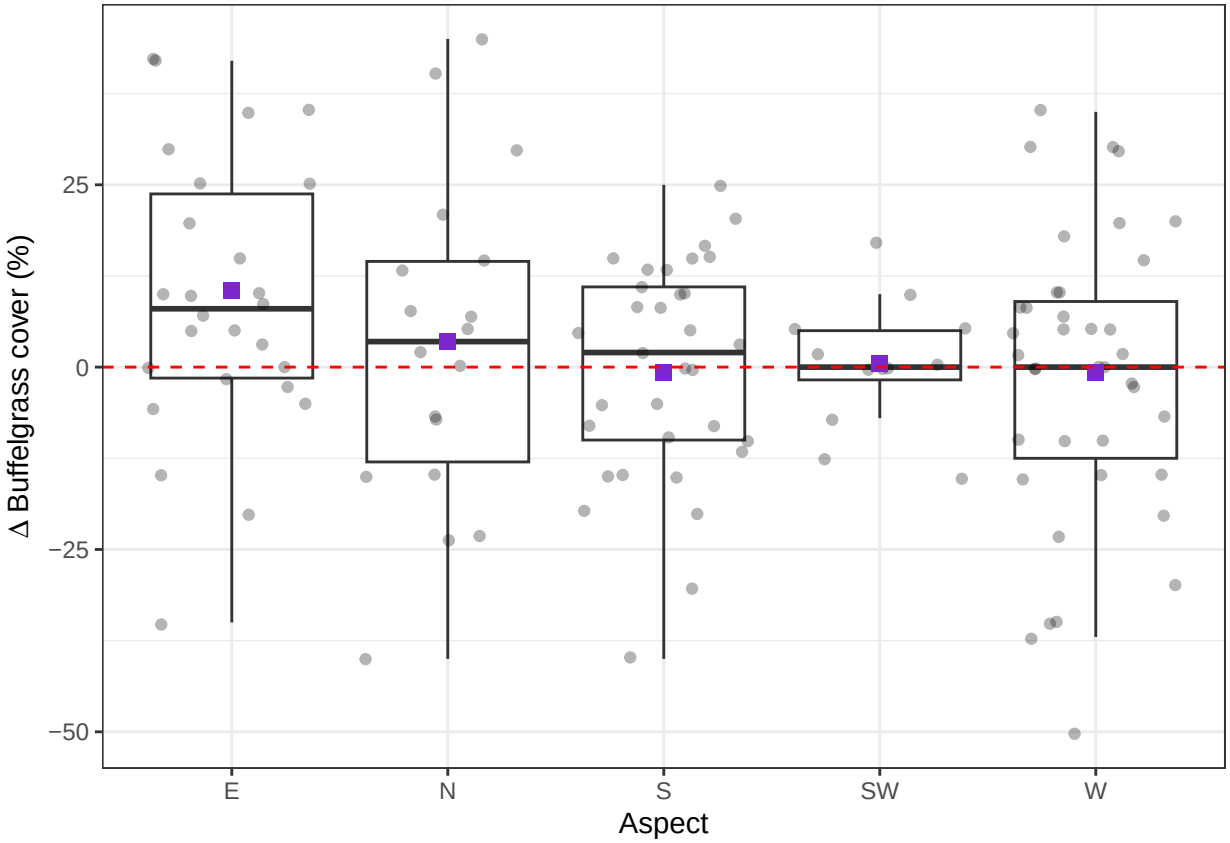

Fig. S9. Effects of aspect on year-to-year change in buffelgrass cover. Purple squares represent model-predicted averages. Horizontal red dashed line indicates no change in buffelgrass cover.

Figure S10

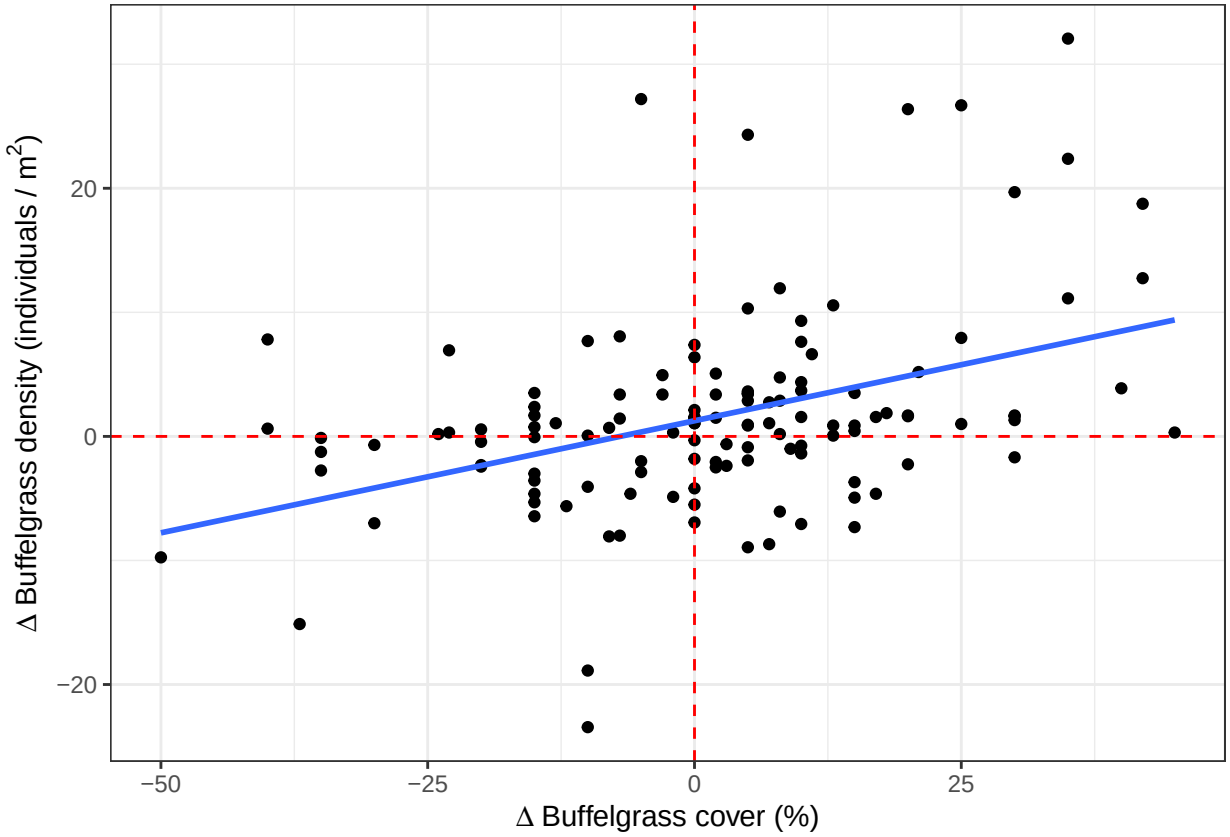

Figure S10. Simple linear regression of year-to-year change in buffelgrass plot density as a function of year-to-year change in buffelgrass plot cover, pooling all sampling locations and times. Vertical red dashed line indicates no change in buffelgrass cover while horizontal line indicates no change in buffelgrass density.

Figure S11

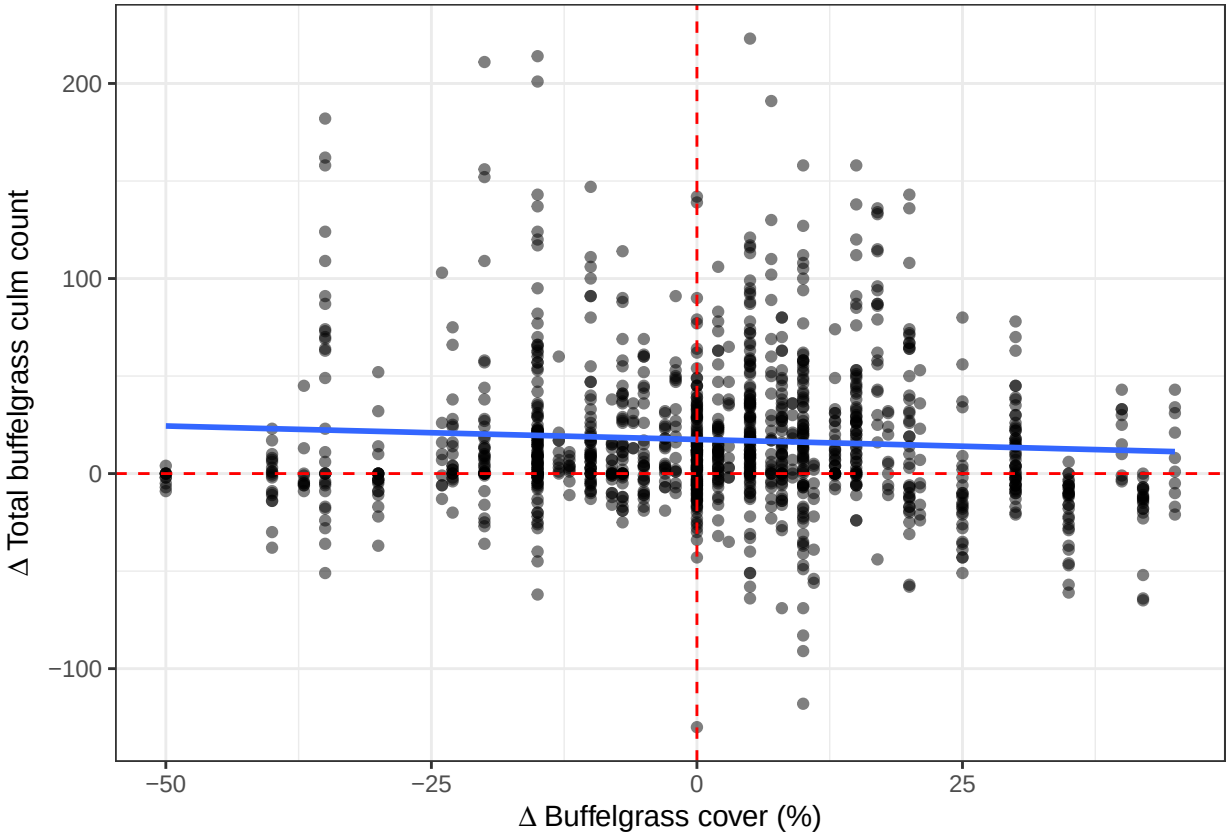

Figure S11. Simple linear regression of year-to-year change in buffelgrass total culm count as a function of year-to-year change in buffelgrass plot cover, pooling all sampling locations and times. Vertical red dashed line indicates no change in buffelgrass cover while horizontal line indicates no change in buffelgrass total culm count.

Figure S12

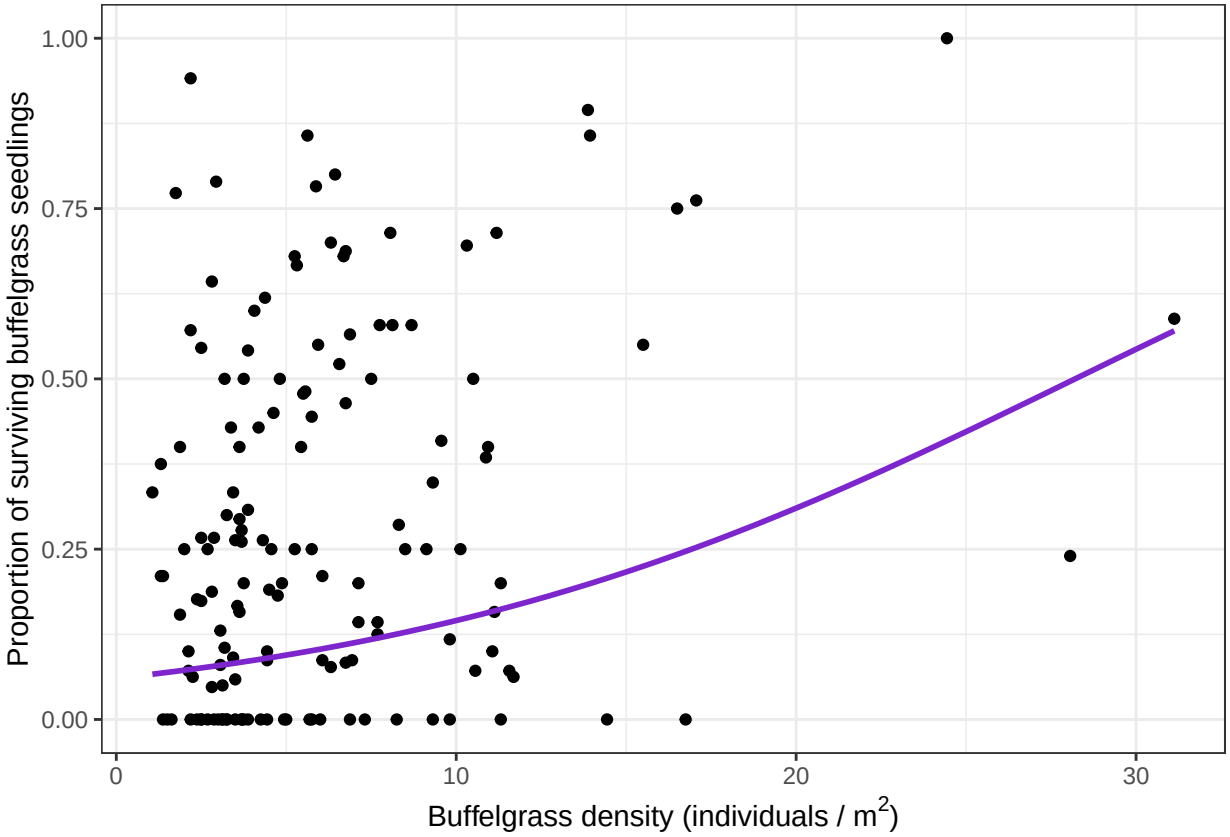

Figure S12. Relationship between buffelgrass seedling survival and buffelgrass density in the plot. Purple line represents model-predicted values.

Figure S13

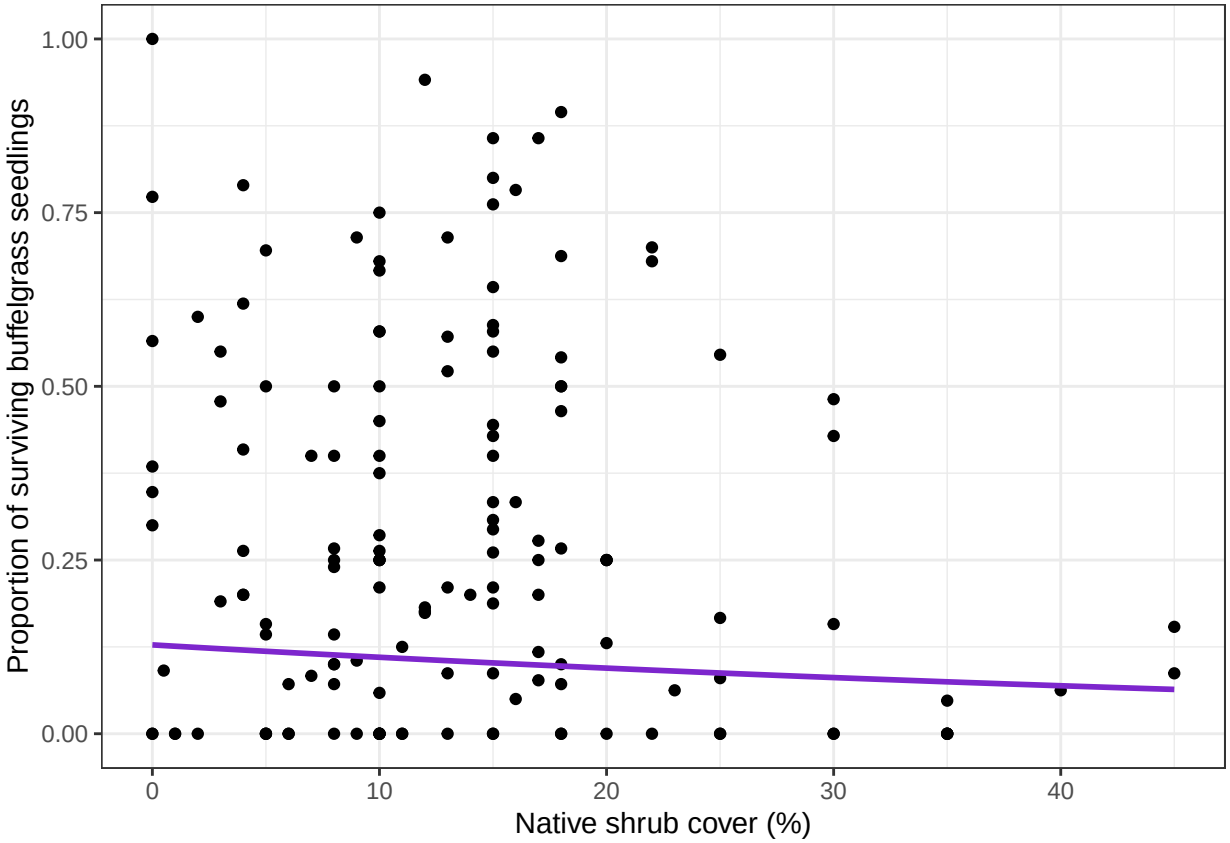

Figure S13. Relationship between buffelgrass seedling survival and native shrub cover in the plot. Purple line represents model-predicted values.